

M61-M66 User M Turn Off (Optional)

The M61 through M66 codes are optional for user interfaces. They turn off one of the relays. Use M51-M56 to turn these on. **[RESET]** turns off all of these relays. See M121-M126 for details on the M-code relays.

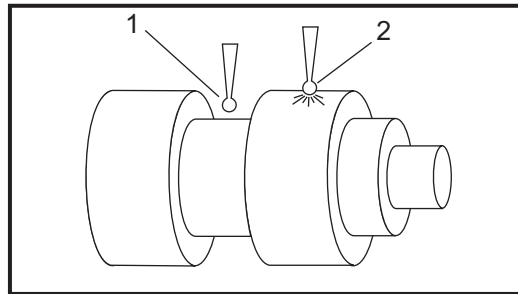
M69 Clear Output Relay

M69 turns off a relay. An example of its usage is M69 P12nnn, where nnn is the number of the relay being turned off. An M69 command can turn off any of the output relays in the range from 12000 to 12255. When using Macros, M69 P12003 does the same thing as using the optional macro command #12003=0, except that it is processed in the same order as axis motion.

M78 / M79 Alarm if Skip Signal Found/Not Found

This M-code is used with a probe. M78 generates an alarm if a programmed skip function (G31) receives a signal from the probe. This is used when a skip signal is not expected, and may indicate a probe crash. M79 generates an alarm if a programmed skip function (G31) did not receive a signal from the probe. This is used when the lack of the skip signal means a probe positioning error. These codes can be placed on the same line as the skip G-code or in any block after.

F8.2: M78/M79 Alarm if Skip Signal Found/Not Found: [1] Signal not found, [2] Signal found.



M85 / M86 Automatic Door Open/Close (Optional)

M85 opens the Auto Door and M86 closes it. The control pendant beeps when the door is in motion.

M88 / M89 High Pressure Coolant On/Off (Optional)

M88 turns on the high pressure coolant option, and M89 turns the coolant off. Use M89 to turn off High Pressure coolant during program execution before rotating the tool turret.



DANGER:

Turn off High Pressure Coolant before performing a tool change.

M95 Sleep Mode

Sleep mode is a long dwell. The format of the M95 command is: M95 (hh:mm).

The comment immediately following M95 must contain the duration, in hours and minutes, that you want the machine to sleep. For example, if the current time were 6 p.m. and you want the machine to sleep until 6:30 a.m. the next day, command M95 (12:30). The line(s) after M95 should be axis moves and spindle warm-up commands.

M96 Jump If No Signal

P - Program block to go to when conditional test is met

Q - Discrete input variable to test (0 to 63)

This code tests a discrete input for 0 (off) status. This is useful for checking the status of automatic work holding or other accessories that generate a signal for the control. The Q value must be in the range 0 to 63, which corresponds to the inputs on the diagnostic display (The upper left input is 0 and the lower right is input 63. When this program block is executed and the input signal specified by Q has a value of 0, the program block Pnnnn is performed (the Pnnnn line must be in the same program).

```
N05 M96 P10 Q8 (Test input #8, Door Switch, until closed) ;
N10 (Start of program loop) ;
. ;
. (Program that machines part) ;
. ;
N85 M21 (Execute an external user function) ;
N90 M96 P10 Q27 (Loop to N10 if spare input [#27] is 0) ;
N95 M30 (If spare input is 1 then end program) ;
```

M97 Local Subprogram Call

This code calls a subprogram (subprogram) referenced by a line number (N) within the same program. A Pnn code is required and must match a line number within the same program. This is useful for subprograms within a program as it does not require a separate program. The subprogram must end with an M99. An Lnn code in the M97 block will repeat the subprogram call nn times.

```
%
O69701 (M97 LOCAL SUBPROGRAM CALL) ;
M97 P1000 L2 (L2 will run the N1000 line twice) ;
M30 ;
N1000 G00 G55 X0 Z0 (N line that will run after M97 P1000 is
run) ;
S500 M03 ;
G00 Z-.5 ;
```

```

G01 X.5 F100. ;
G03 ZI-.5 ;
G01 X0 ;
Z1. F50. ;
G28 U0 ;
G28 W0 ;
M99 ;
%
```

M98 Subprogram Call

P - The subprogram number to run
L - Repeats the subprogram call (1-99) times.
(<PATH>) - The Subprogram's directory path

M98 calls a subprogram in the format M98 Pnnnn, where Pnnnn is the number of the program to call, or M98 (<path>/Onnnnn), where <path> is the device path that leads to the subprogram.

The subprogram must contain an M99 to return to the main program. You can add an Lnn count to the M98 block M98 to call the subprogram nn times before continuing to the next block.

When your program calls an M98 subprogram, the control looks for the subprogram in the main program's directory. If the control cannot find the subprogram, it then looks in the location specified in Setting 251. Refer to page **170** for more information. An alarm occurs if the control cannot find the subprogram.

M98 Example:

The subprogram is a separate program (O00100) from the main program (O00002).

```

%
O00002 (PROGRAM NUMBER CALL);
M98 P100 L4 (CALLS O00100 SUB 4 TIMES) ;
M30 ;
%
%
O00100 (SUBPROGRAM);
M00 ;
M99 (RETURN TO MAIN PROGRAM) ;
%

%
O00002 (PATH CALL);
M98 (USB0/O00001.nc) L4 (CALLS O00100 SUB 4 TIMES) ;
```

```

M30 ;
%
%
O00100 (SUBPROGRAM) ;
M00 ;
M99 (RETURN TO MAIN PROGRAM) ;
%
```

M99 Subprogram Return or Loop

This code has three main uses:

1. An **M99** is used at the end of a subprogram, local subprogram, or macro to return back to the main program.
2. An **M99 Pnn** jumps the program to the corresponding **Nnn** in the program.
3. An **M99** in the main program causes the program to loop back to the beginning and run until **[RESET]** is pressed.

Programming Notes - You can simulate Fanuc behavior by using the following code:

	Haas	Fanuc
Calling program:	O0001	O0001
	...	...
	N50 M98 P2	N50 M98 P2
	N51 M99 P100	...
	...	N100 (continue here)
	N100 (continue here)	...
	...	M30
	M30	
Subprogram:	O0002	O0002
	M99	M99 P100

M99 With Macros - If the machine is equipped with the optional macros, you can use a global variable and specify a block to jump to by adding `#nnnnn = dddd` in the subprogram and then using `M99 P#nnnnn` after the subprogram call.

M104 / M105 Probe Arm Extend/Retract (Optional)

The optional tool setting probe arm is extended and retracted using these M-codes.

M109 Interactive User Input

P - A number in the range (500-549) representing the macro variable of the same name.

This M code allows a G-code program to place a short prompt (message) on the screen. A macro variable in the range 500 through 549 must be specified by a **P** code. The program can check for any character that can be entered from the keyboard by comparing with the decimal equivalent of the ASCII character.

T8.1: Values for ASCII Characters

32		space	59	;	semicolon
33	!	exclamation mark	60	<	less than
34	"	double quotation mark	61	=	equals
35	#	number sign	62	>	greater than
36	\$	dollar sign	63	?	question mark
37	%	percent sign	64	@	at sign
38	&	ampersand	65-90	A-Z	capitol letters
39	'	closed single quote	91	[	open square bracket
40	(	open parenthesis	92	\	backslash
41	)	close parenthesis	93	]	closed square bracket
42	*	asterisk	94	^	carrot
43	+	plus sign	95	_	underscore
44	,	comma	96	'	open single quote
45	-	minus sign	97-122	a-z	lowercase letters
46	.	period	123	{	open curly bracket
47	/	slash	124		vertical bar

48-57	0-9	numbers	125	}	closed curly bracket
58	:	colon	126	~	tilde

The following sample program asks the user a Yes or No question, then wait for either a Y or an N to be entered. All other characters are ignored.

```
%
o61091 (57 M109_01 Interactive User Input) ;
N1 #501= 0. (Clear the variable) ;
N5 M109 P501 (Sleep 1 min?) ;
IF [ #501 EQ 0. ] GOTO5 (Wait for a key) ;
IF [ #501 EQ 89. ] GOTO10 (Y) ;
IF [ #501 EQ 78. ] GOTO20 (N) ;
GOTO1 (Keep checking) ;
N10 (A Y was entered) ;
M95 (00:01) ;
GOTO30 ;
N20 (An N was entered) ;
G04 P1. (Do nothing for 1 second) ;
N30 (Stop) ;
M30 ;
%
```

The following sample program asks the user to select a number, then wait for a 1, 2, 3, 4 or a 5 to be entered; all other characters are ignored.

```
%
O61092 (58 M109_02 Interactive User Input) ;
N1 #501= 0 (Clear Variable #501) ;
(Variable #501 will be checked) ;
(Operator enters one of the following selections) ;
N5 M109 P501 (1,2,3,4,5) ;
IF [ #501 EQ 0 ] GOTO5 ;
(Wait for keyboard entry loop until entry) ;
(Decimal equivalent from 49-53 represent 1-5) ;
IF [ #501 EQ 49 ] GOTO10 (1 was entered go to N10) ;
IF [ #501 EQ 50 ] GOTO20 (2 was entered go to N20) ;
IF [ #501 EQ 51 ] GOTO30 (3 was entered go to N30) ;
IF [ #501 EQ 52 ] GOTO40 (4 was entered go to N40) ;
IF [ #501 EQ 53 ] GOTO50 (5 was entered go to N50) ;
GOTO1 (Keep checking for user input loop until found) ;
N10 ;
(If 1 was entered run this sub-routine) ;
```

```

      (Go to sleep for 10 minutes) ;
#3006= 25 (Cycle start sleeps for 10 minutes) ;
M95 (00:10) ;
GOTO100 ;
N20 ;
      (If 2 was entered run this sub routine) ;
      (Programmed message) ;
#3006= 25 (Programmed message cycle start) ;
GOTO100 ;
N30 ;
      (If 3 was entered run this sub routine) ;
      (Run sub program 20) ;
#3006= 25 (Cycle start program 20 will run) ;
G65 P20 (Call sub-program 20) ;
GOTO100 ;
N40 ;
      (If 4 was entered run this sub routine) ;
      (Run sub program 22) ;
#3006= 25 (Cycle start program 22 will be run) ;
M98 P22 (Call sub program 22) ;
GOTO100 ;
N50 ;
      (If 5 was entered run this sub-routine) ;
      (Programmed message) ;
#3006= 25 (Reset or cycle start will turn power off) ;
#1106= 1 ;
N100 ;
M30 ;
%
```

M110 / M111 Secondary Spindle Chuck Clamp/Unclamp (Optional)

These M codes will clamp and unclamp the secondary spindle chuck. OD / ID clamping is set with Setting 122.

M112 / M113 Secondary Spindle Air Blast On/Off (Optional)

M112 turns on the secondary spindle air blast. M113 turns the secondary spindle air blast off. M112 Srrr Pnnn (rrr is in RPM and nnn is in milliseconds) turns the air blast on for the specified time, rotates the spindle at the specified speed while the air blast is on, then turns off both the spindle and the air blast automatically.

M114 / M115 Secondary Spindle Brake On/Off (Optional)

M114 applies a caliper-style brake to hold the secondary spindle, while M115 releases the brake.

M119 Secondary Spindle Orient (Optional)

This command orients the secondary spindle (DS lathes) to the zero position. A **P** or **R** value is added to position the spindle to a specific position. A **P** value positions the spindle to that whole degree (e.g. **P120** is 120°). An **R** value positions the spindle to a fraction of a degree (e.g. **R12.25** is 12.25°). The format is: M119 **P**xxx/M119 **R**xx.x. The spindle angle is viewed in the Current Commands Tool Load screen.

M121-M128 User M-codes (Optional)

The M121 through M128 codes are optional for user interfaces. They activate relays 0 through 255, wait for the M-fin signal, release the relay, and wait for the M-fin signal to cease. **[RESET]** terminates any operation that is hung-up waiting for M-fin.

M129 Set Output Relay with M-Fin

P - Discrete output relay from 0 to 255.

M129 turns on a relay, pauses the program, and waits for an external M-Fin signal. An example of its usage is M129 **P**nnn, where nnn is the relay number being turned on. When the control receives the M-Fin signal, the relay turns off and the program continues. **[RESET]** stops any operation waiting for a relay-activated accessory to finish.

M130 Display Media / M131 Cancel Display Media

M130 Lets you display video and still images during program execution. Some examples of how you can use this feature are:

- Providing visual cues or work instructions during program operation
- Providing images to aid part inspection at certain points in a program
- Demonstrating procedures with video

The correct command format is M130 (file.xxx), where file.xxx is the name of the file, plus the path, if necessary. You can also add a second comment in parentheses to appear as a comment at the top of the media window.



NOTE:

M130 uses the subprogram search settings, Settings 251 and 252 in the same way that M98 does. You can also use the Insert Media File command in the editor to easily insert an M130 code that includes the filepath. Refer to page 122 for more information.

Permitted file formats are MP4, MOV, PNG, and JPEG.



NOTE:

For the fastest loading times, use files with pixel dimensions divisible by 8 (most unedited digital images have these dimensions by default), and a maximum pixel size of 1920 x 1080.

Your media appears in the Media tab under Current Commands. The media displays until the next **M130** displays a different file, or **M131** clears the media tab contents.

F8.3: Media Display Example - Work Instruction during a Program



M133 / M134 / M135 Live Tool Fwd/Rev/Stop (Optional)

M133 turns the live tool spindle in the forward direction. **M134** turns the live tool spindle in the reverse direction. **M135** stops the live tool spindle.

Spindle speed is controlled with a **P** address code. For example, **P1200** would command a spindle speed of 1200 RPM.

M138 / M139 Spindle Speed Variation On/Off

Spindle Speed Variation (SSV) lets you specify a range within which the spindle speed continuously varies. This is helpful in suppressing tool chatter, which can lead to an undesirable part finish and/or damage to the cutting tool. The control varies the spindle speed based on Settings 165 and 166. For example, to vary spindle speed +/- 100 RPM from its current commanded speed with a duty cycle of 1 seconds, set Setting 165 to 100 and Setting 166 to 1.

The variation you use depends on the material, tooling, and the characteristics of your application, but 100 RPM over 1 second is a good starting point.

You can override the values of settings 165 and 166 using P and E address codes when used with M138. Where P is SSV Variation (RPM) and E is the SSV Cycle (Sec). See example below:

```
M138 P500 E1.5 (Turn SSV On, vary the speed by 500 RPM, cycle every 1.5 seconds);
```

```
M138 P500 (Turn SSV on, vary the speed by 500, cycle based on setting 166);
```

```
M138 E1.5 (Turn SSV on, vary the speed by setting 165, cycle every 1.5 seconds);
```

M138 is independent of spindle commands; once commanded, it is active even when the spindle is not turning. Also, M138 remains active until canceled with M139, or at M30, Reset, or Emergency Stop.

M143 / M144 / M145 Secondary Spindle Fwd/Rev/Stop (Optional)

M143 turns the secondary spindle in the forward direction. M144 turns the secondary spindle in the reverse direction. M145 stops the secondary spindle

The subspindle speed is controlled with an P address code, for example, P1200 commands a spindle speed of 1200 RPM.

M146 Steady Rest Clamp / M147 Steady Rest Unclamp

M146 clamps the steady rest and M147 unclamps it.

M154 / M155 C-Axis Engage/Disengage (Optional)

This M code is used to engage or disengage the optional C-Axis motor.